Jeevan John Jacob

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Summary

Data Science graduate student skilled in AI, machine learning, and full-stack development. Experienced in building real-time intelligent systems and impactful research across security, healthcare, and IoT. Strong problem-solver and communicator, passionate about leveraging technology for real-world impact.

Education

Amrita School of Computing — 2024 – Present
M.Tech in Data Science — *CGPA: 8.25*

Amrita School of Engineering, Bangalore — 2020 – 2024
B.Tech in Electronics and Communications Engineering — *CGPA: 7.02*

Sri Chaitanya P U College, Bangalore — 2018
12th Grade Education — *85%*

Experience

GE Healthcare – Patient Care Solutions Technology — Oct 2025 – Present
Software Engineer Intern — *Bengaluru, India*

- Migration of embedded medical device software from 32-bit IMX6 to 64-bit IMX95 architecture for next-generation hardware platforms.
- Developed and maintained Qt-based frontend systems for device user interfaces.
- Supported development and testing of microservices-based product software components.

Projects

Real-Time Fire Detection & Extinguishing System | *Python, YOLOv5, OpenCV, Streamlit, Twilio API*

- Built a real-time detection system for fire, smoke, and flame from webcam feeds using YOLOv5 and OpenCV; achieved mAP of 0.456 and 60%+ precision on a custom Roboflow dataset.
- Built Streamlit dashboard for visualization, sprinkler simulation, and manual override; integrated Twilio API for WhatsApp alerts and event logging with analytics.

Harmful Brain Activity Classification | *Python, EEG, Machine Learning*

- Classified six types of harmful brain activity using EEG data from 17,300 patients; reduced features from 224 to 52 using PCA while maintaining 94%+ accuracy.
- Achieved 95% accuracy with Random Forest, outperforming all baseline models; explored deep learning and GANs for future improvements.

Progressive Reading Aid for Visually Impaired | *Raspberry Pi, Deep Learning, Object Detection*

- Created a real-time reading aid using object detection on Raspberry Pi to assist visually impaired users in reading and navigating their environment.

Technical Skills

Programming: Python, C++, QML, MATLAB, MySQL, PowerBI, Excel, Yocto Linux
ML / AI: Scikit-learn, YOLOv5, OpenCV, TensorFlow, Algorithms, Deep Learning, Data Analysis
Hardware & Tools: Arduino, Simulink, Raspberry Pi, Qt Creator, Ethereum, Yocto Project, MongoDB Compass
Languages: English, Hindi, Kannada, Malayalam, Tamil

Publications

[1] Deepthi, Y. P., Pranav Kalaga, Santosh Kumar Sahu, **Jeevan John Jacob**, Kiran P. S, and Quanjin Ma. "AI-based machine learning prediction for optimization of copper coating process on graphite powder for green composite fabrication." *International Journal on Interactive Design and Manufacturing (IJIDeM)* 19, no. 6 (2025): 4123–4130.

[2] **Jacob, Jeevan John**, P. S. Kiran, Jaigovind B. Shenoy, and Sagar Basavaraju. "Enhancing Security: A Hybrid Encryption Algorithm for Effective Key Generation." In *2024 2nd International Conference on Recent Trends in Microelectronics, Automation, Computing and Communications Systems (ICMACC)*, pp. 652–657. IEEE, 2024.

[3] Shenoy, Jaigovind B., P. S. Kiran, **Jeevan John Jacob**, and V. Bhavana. "Performance Analysis of Vehicular Networks with Prioritized Traffic." In *2024 15th International Conference on Computing Communication and Networking Technologies (ICCCNT)*, pp. 1–7. IEEE, 2024.

Courses & Certifications

Deep Learning Onramp – MathWorks (Feb 2024): Image classification using pretrained deep neural networks in MATLAB.

Google Cloud Certifications – Google (May 2022): BigQuery ML, ML APIs, Explainable AI for data engineering and predictive modeling.

Online Presence

GitHub: github.com/Jeevanjohnjacob
YouTube: youtube.com/@techwithtriplej8094